

Automatic Oil Palm Unstripped Bunch (USB) Counting System based on Faster RCNN and Object Tracking

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Abstract— USB Palm Oil Counting (Unstripped Bunch) is important in oil palm processing mills. Information on the number of USBs is essential because it shows the level of efficiency of the palm oil processing plant. Due to its complexity, palm oil mills rely solely on manual calculations, which is inefficient workforce waste. Challenging aspects such as partial occlusion, overlap, and even different perspectives limit the use of traditional computer vision techniques. In recent years, deep learning has become increasingly popular for computer vision applications due to its superior performance over conventional methods. This paper proposes a deep learning solution to solve USB computing problems in palm oil mills. Our proposed automated USB counter system consists of an object detector built on the RCNN Faster architecture and an object tracker made in the euclidean distance. Our proposed system identifies and counts USBs with an average accuracy of 71.5% in testing.

Keywords—automatic, unstripped bunch, counting, faster r-cnn, object, tracker

I. INTRODUCTION

The processing of oil palm fruit into CPO is a complex process. Because the oil palm production process is complicated, there will be losses. Losses in oil palm mills include losses caused by the poor implementation of the sterilization process and the extraction of oil palm fruit from bunches[1].

Improper sterilization and seeding process of oil palm fruit bunch will result in empty bunches containing oil palm fruit. This bunch is called Unstripped Bunch or USB. In Malaysia, loss in USB, estimated reaches an average of 0.05% / TBS[2]. In one case study, in Indonesia (another major CPO producer), loss caused by USB is a bit bigger, 2.2%[3]. In some cases where the bunches are not treated properly, USB loss could reach 40% / TBS[4].

Based on this, we can conclude, the performance of the oil palm mills can be seen based on small USB losses; in other words, the performance measurement of oil palm mills can be seen based on USB loss; unfortunately, no method can automatically count the USB. So far, the USB surveillance has been done manually and is neglected in some cases[5].

Object counting is a major task in computer vision and has been widely applied to crowd monitoring[6] [7] [8] [9] [10] [11] [12] [13], blood cell counting[14] [15] [16], and fruits detection[17] [18] [19]. Initially, the counting task was carried out through object detection. However, later mainstream methods involved density map regression. Although object counting is still a challenge because images have various appearances, scale changes, and object distributions are not homogeneous, because of its extensive application, automatic

object counter remains an area that attracts researchers to develop its application or method.

In the production process of an industry, it is often necessary to calculate objects that are run on conveyors. This problem is usually solved using sensors or electronic transducers such as photo or laser sensors instead of computer vision[20]. However, along with the development of computer vision technology and driven by unique conditions, several researchers have tried to develop counting objects on a conveyor using image processing[21].

The counting of a USB on a running conveyor requires a different solution. Attempts to count objects on a conveyor belt as carried out by Vanijirattikhan et al. cannot be carried out for this USB counting. The redundancy count and line camera methods do not detect objects of more than one class on the conveyor. In this case, there are two objects, namely EFB and USB, and these two objects must be distinguished. Another problem faced by the direct counting method is the occurrence of double-counting, where the same object is counted twice. The researcher solves this problem by calculating lines that form the calculated area. This research project will use an object tracking approach to solve this problem.

This research aims to develop a UB counting system by utilizing faster R-CNN as an object detector and a tracking method based on Euclidean distance to reduce double counting.

II. RESEARCH PROGRESS

A. USB (Unstripped Bunch)

USB (Unstripped Bunch) loose definition is an empty bunch with fruitlets still attached. Fig. 1 shows a sample of fruitlets on a USB.



Fig. 1. A USB with fruitlets (red box) attached to it

According to Adzmi et al., USB is an empty bunch with more than 20 fruitlets still attached. In palm oil mills, USB is observed at the EFB (Empty Fruit Bunch) conveyor. EFB is a bunch without fruitlets attached to it, as depicted in Fig. 2.



Fig. 2. An EFB (Empty Fruit Bunch)

B. Object detection and Object Counting With Deep Learning

Researchers have exploited the profound learning ability to recognize and detect objects. Its application is extensive in agriculture, health, transportation, medical, and even defense. Deep learning has proven to be the most valuable solution for object detection in different circumstances.

Object learning systems based on deep learning are widely used in almost all fields of agriculture, such as object identification[22], object classification [12] [13] [25], and object detection and counting [26] [18].

Several researchers have developed a deep learning approach for object counting in agriculture. The detection and counting of tomatoes were carried out by Afonso et al. In his study, Afonso et al. succeeded in developing a tomatoes detection and counting system using deep learning. Tian et al. and Cowton et al. extended an automatic pig counter for pig farming[27]. This counting system is also based on deep knowledge. However, deep learning-based object detection application used to calculate USB in palm oil production has never been implemented. In this paper, we propose a running USB counter based on the faster RCNN-based USB detection.

III. OBJECT COUNTING WITH FASTER RCNN

Fig. 3 shows a proposed block diagram USB counter system based on Faster RCNN. The main components of this USB counter system are the USB detector component and the USB counter component.

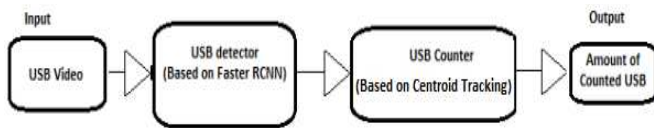


Fig. 3. USB Counting System Block Diagram

We propose fruitlets detection and bunch detection for detecting USB. The USB detector backbone is built from Faster RCNN. The output for object detection using a faster RCNN is in the form of two parameters. The first parameter is a classification parameter, while the second is a bounding box parameter, a regression parameter. We will take the bounding box parameter for the USB counter to calculate the USB centroid and check whether the USB is an old one or a new one.

Based on the basic definition that USB is EFB which still has a fruitlet in it, to justify a sign is USB or not based on the presence of a fruitlet in the bunch. In labeling, we propose detecting one fruitlet, two fruitlets, three fruitlets, and four

fruitlets. This selection is based on the consideration that in factory operations, the definition of USB is often different in each palm oil mills company. A bunch is said to be a USB if inside the bunch there are fruitlets. Based on this definition, the steps are taken to determine whether a bunch is a USB bunch or not; the first step is to get the BBox coordinates $(X1_{usb}, Y1_{usb}, X2_{usb}, Y2_{usb})$ of the bunch and then get the fruitlet coordinates $(X1_{fruitlet}, Y1_{fruitlet}, X2_{fruitlet}, Y2_{fruitlet})$. If the fruit coordinates are in the bbox coordinates of the bunches, it is concluded that the detected bunches are USB. The pseudo-code for determining a bunch is USB is as follows:

```

Procedure get USB ()
begin
   $(X1_{usb}, Y1_{usb}, X2_{usb}, Y2_{usb}) \leftarrow$  Read USB bounding box
  If (label = fruitlet1 or fruitlet2 or fruitlet3 or fruitlet 4 )
     $(X1_{fruitlet}, Y1_{fruitlet}, X2_{fruitlet}, Y2_{fruitlet}) \leftarrow$  read fruitlets BBox
    If  $X1 < X2$ , then
      USB = true
  End procedure

```

A complete flowchart for the proposed USB counter system is shown in Fig. 4.

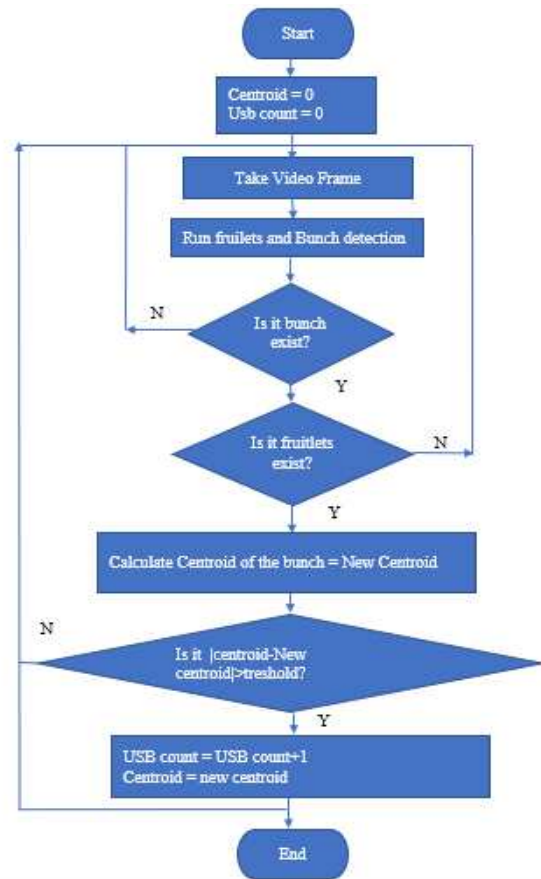


Fig. 4. Workflow for USB counter

A. Custom datasets

The labeling for custom datasets used in this study is the single fruitlets, double fruitlets, triple fruitlets, quarter fruitlets, and bunch. The images of the datasets are captured from video surveillance mounted on top of the exhaust conveyor of a palm oil mill. This video is taken from PT. Sawit Arum Madani, a palm oil mill located in Blitar, Indonesia in May, 2019. The number of images available is

500. Four hundred sixty images (92%) were used as train data, and 40 (8%) images were used as test data. Fig. 5-9 show examples of single fruitlets, double fruitlets, triple fruitlets, quarter fruitlets, and bunch, respectively. Words 'one,' 'two,' 'three,' 'four,' and 'bunch' are used to label single, double, triple, quartet, and bunch labeling, and for labeling LabelImg [28] is used for labeling images.



Fig. 5. Single fruitlet



Fig. 6. Double fruitlet



Fig. 7. Triple fruitlet

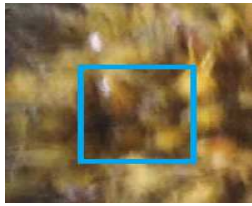


Fig. 8. Quartet fruitlet



Fig. 9. Oil palm bunch

B. USB Counter based on centroid tracking

An EFB is considered a USB in our USB counting system if it has at least one fruitlets. Hence the first step that is done is fruitlet detection and bunch detection. After the USB has been successfully detected, we use an object tracking method to determine whether the USB detected earlier is a new USB or the USB in the previous frame. The algorithm that we propose for USB tracking is the centroid tracking algorithm. The object tracking algorithm works based on the Euclidean distance between the existing object's centroid (i.e., the object that the centroid tracer has seen in the previous video frame) and the center of the new object between subsequent frames in the video as depicted in Fig. 10.



Fig. 10. Two USB with its centroid and Euclidean Distance,

Ct = Centroid, ED = euclidean distance

The two centroids are compared, and if they exceed a certain threshold, then the USB that appears in this new frame is considered the new USB. It is done based on the assumption that the centroid of the same object in the old frame and the new frame do not differ much. The purpose of using the calculation method by utilizing object tracking is to reduce the double-counting of objects.

IV. RESULT

We conducted two tests for performance evaluation of the RCNN Faster-based USB counter system. The first test measures object detector capabilities to recognize trained labels (single fruitlets, double fruitlets, triple fruitlets, and bunch). The second test measures USB counter capabilities to detect and count USB in video streaming.

A. The testing results of the Faster R-CNN ability to detect oil palm fruitlets and fruit bunches

Test results of detector's ability to identify objects being trained. This test is carried out on a datasheet containing ten images. Each image has fruitlets on it.

1) Test on a single fruitlet

Fig.11-12 shows the sample test results of the detector's ability to recognize a single fruitlet on one USB. The target image is depicted in Fig.11. The tests carried out using 10 test images showed that the system's ability to detect single fruit on USB had an average accuracy of 79.6%.



Fig. 11. Single fruitlet target



Fig. 12. The single fruitlet detection result

2) Test on double fruitlets

The example of test results for the detection of double fruitlets is shown in Fig.13-14. The target fruitlets are marked with a circle. The results of the double fruitlets test with a total of 10 test images showed that the fruitlets detection system could recognize double fruitlets with an accuracy of 70%.



Fig. 13. Double fruitlet target



Fig. 14. The double fruitlet detection result

The results of the double fruitlets test with a total of 10 test images showed that the fruitlets detection system could recognize double fruitlets with an accuracy of 70%.

3) Test on triplet fruitlets

The example of target image for triplet fruitlets detection testing is depicted in Fig. 15, and the detection result depicted in Fig. 16



Fig. 15. Triplet fruitlet target



Fig. 16. Triplet fruitlet result detection result

The test results for all test images (a total of 10 images) show that the system can recognize triplet fruits with an accuracy of 84.5%.

4) Test on quartet fruitlets

As with the previous test, for testing the quartet fruitlets, 10 test images were used. Figure 17 is a sample of the intended test. In Fig.17, the target is quarter fruitlets marked with a circle; the test image has two quartet fruitlets. The detection result is shown in Fig. 18, where the system could detect one of two quartet fruitlets (accuracy at 50%).



Fig. 17. Quartet fruitlet target



Fig. 18. Quartet fruitlet detection result

Based on the test results obtained data that the accuracy of the system in detecting the quartet of fruitlets reached 84.9%

5) Test on oil palm bunch detection

Tests were also carried out on the system's ability to detect the presence of oil palm bunches. The bunches that were detected were the ones that had fruitlets inside (USB) and bunches that did not have fruitlets (EFB). Fig. 19 depicts an image with two oil palm bunch (marked red circle), and Fig. 20 shows the bunch detection result. In this case, all two bunches were successfully detected.



Fig. 19. Oil palm bunch target

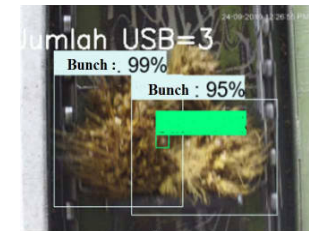


Fig. 20. The result of oil palm bunch detection

We conduct bunch detection for ten target images, and the mean accuracy of oil palm bunch detection is 95%.

Based on the five experimental data above, we can conclude that this object detection system based on Faster RCNN is capable recognize all trained labels on images tested with mean accuracy is 82.8%.

That is means an object detection system based on faster RCNN could be used as a USB or Unstripped Bunch counter in palm oil mills.

A. Testing on USB counter software prototype

USB software prototype tested on three different video files. Each of the files has a different amount of USB and EFB (this number calculated manually) with a duration time of 5 minutes. The USB counter has a user interface as a human-machine interface, as depicted in Fig. 21.



Fig. 21. USB counting software user interface

Users must select the video that wants to be counted. After the user selects the video, run the program using click the start button, and the program will count the USB. The result is stored in a .xls file with 'Jumlah_USB.xls' as depicted in Fig.22.

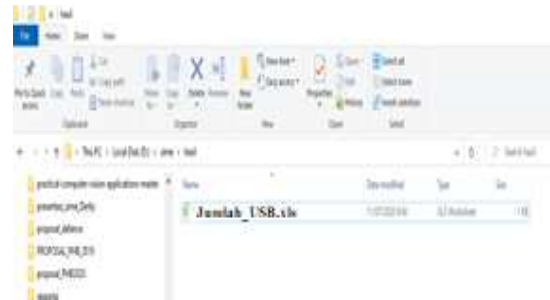


Fig. 22. Folder for storage the test result

For testing the USB Counter, we run it to count USB in three different as aforementioned, and the result is shown in table 6.

TABLE I. TOTAL NUMBER OF OIL PALM BUNCHES IN THE VIDEO BASED ON MANUAL AND AUTOMATIC COUNTING

No.	File Name	Number of the total bunch (USB and EFB) in video		
		Manually counted	USB counter (USB+EFB)	Accuracy
1	Test1	78	63	80%
2	Test2	90	60	66.7%
3	Test3	95	81	85.5%

TABLE II. USBs COUNTING RESULTS ON A VIDEO FILE

No.	File Name	Number of USBs in video		
		Manually Counted	USB counter	Accuracy

1	Test1	75	58	77.3%
2	Test2	81	50	61.7%
3	Test3	83	63	75.9%

The means accuracy USB counter for oil palm bunch counting is 77%, and the average accuracy for USB counting is 71.5%.

V. CONCLUSION

Based on performance, we successfully proposed a deep learning-based USB detection and counter system using centroid tracking as part of the counter system with an average accuracy of 71.5%. This shows that the faster RCNN-based object detection system can be used as the backbone of the automatic counting system. The Euclidean-based tracking method can be used to avoid recounting objects that have been counted.

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